Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Machine Learning

> Image Classification

Setting up ML server

- Get code
- `pnpm install`
- `pnpm run build`
- `pnpm run start`
- Test if server is running.
  - http://localhost:3003

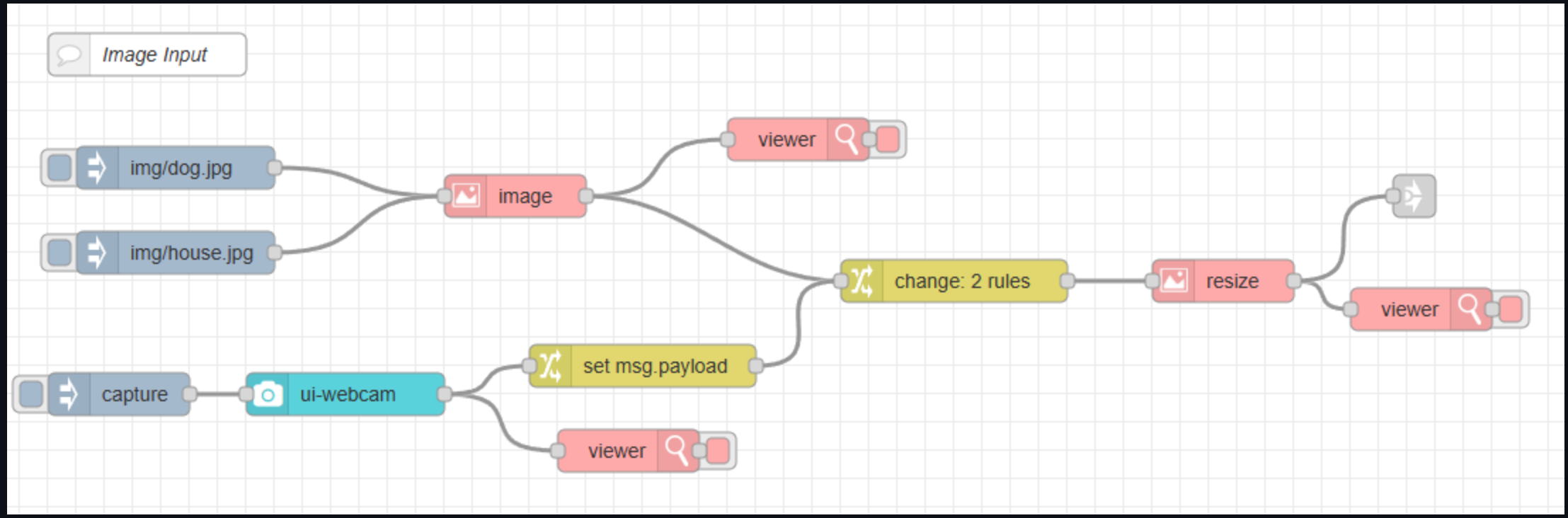
Install Additional Modules

- Go to Menu (top right) -> Manage palette -> Install
- Install the following modules:
  - `node-red-contrib-image-tools`
  - `@sumit_shinde_84/node-red-dashboard-2-ui-webcam`

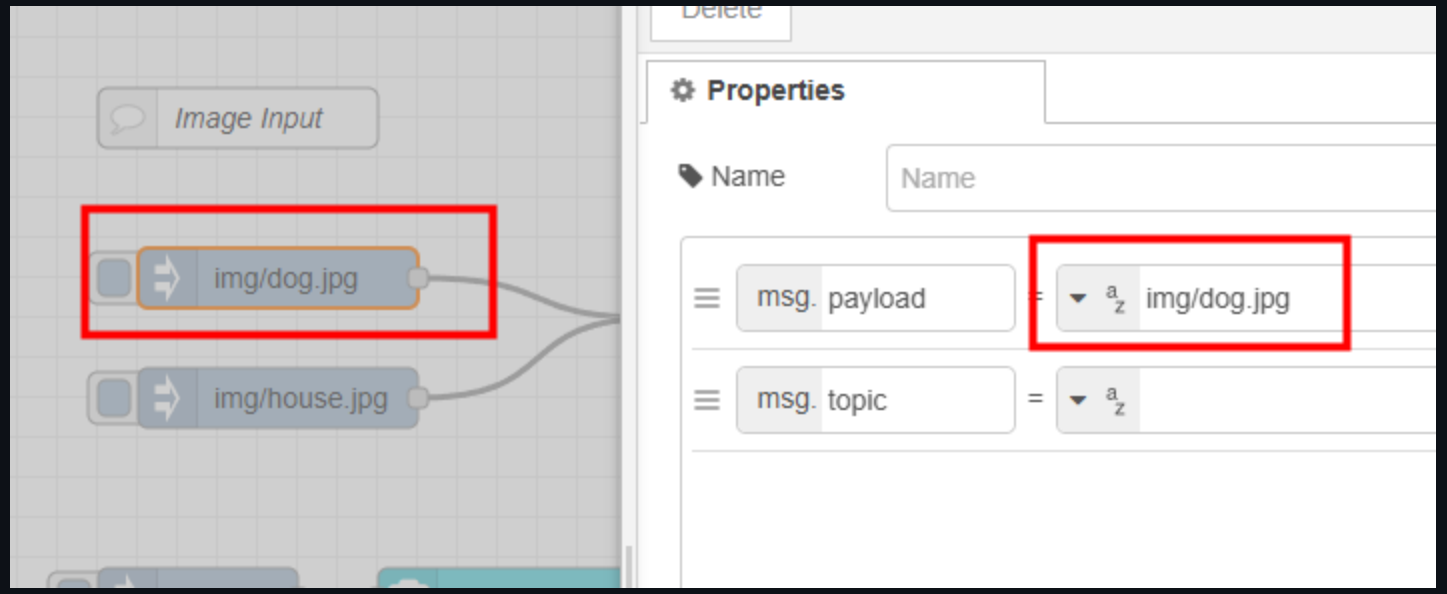
Prepare Mock Images

- Copy image from `ml-express/temp/img` to your node-red installation directory under `img` folder.

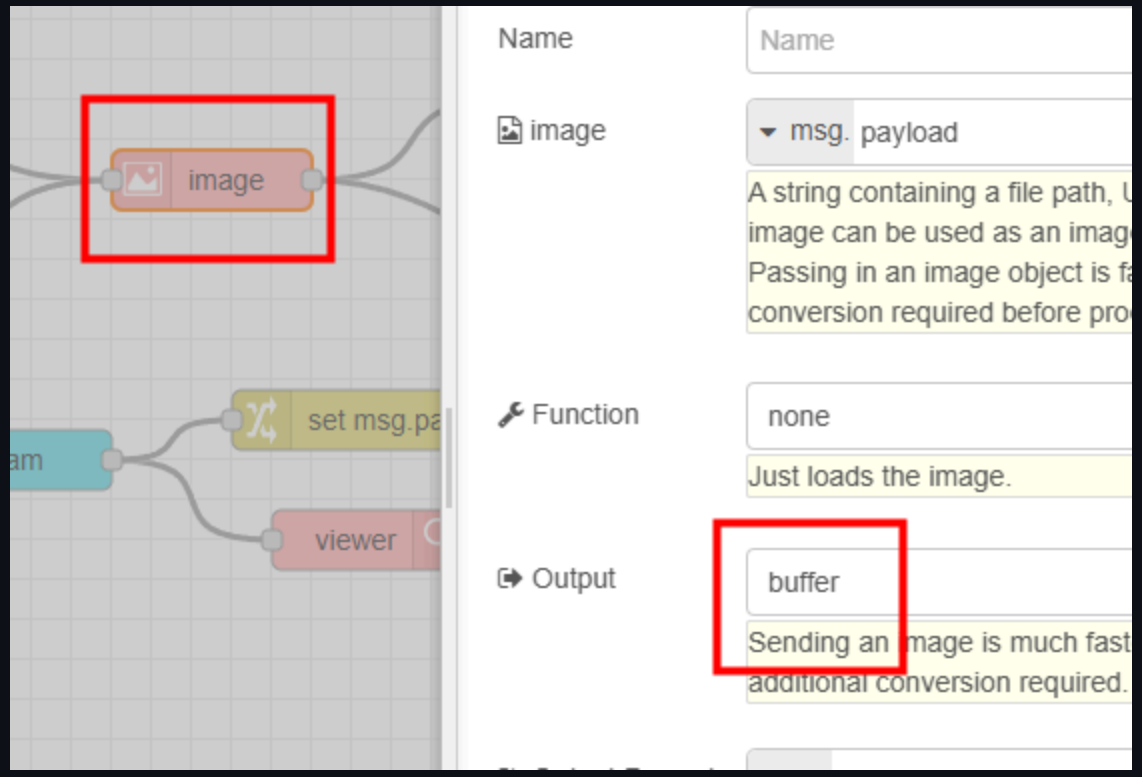
Image input flow



`Inject` Node Configuration

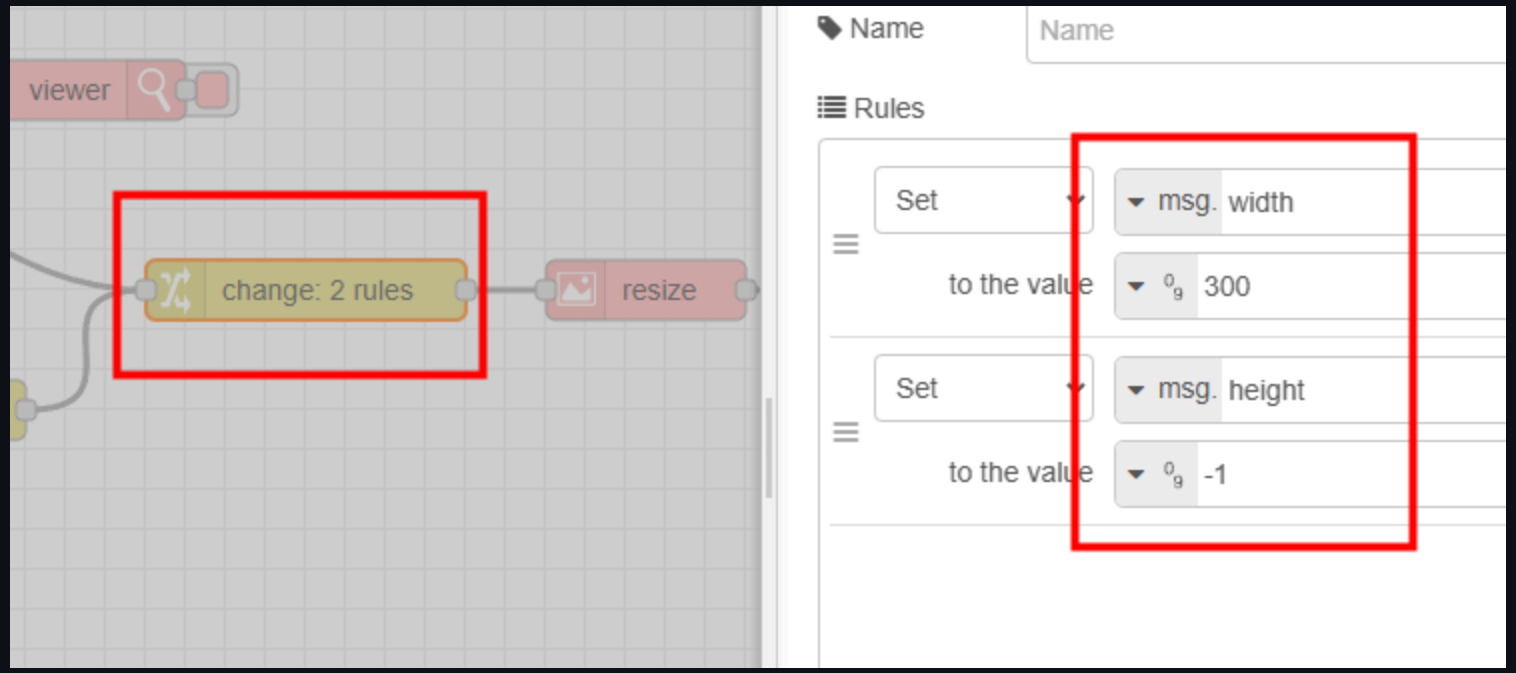


`Image` Node Configuration



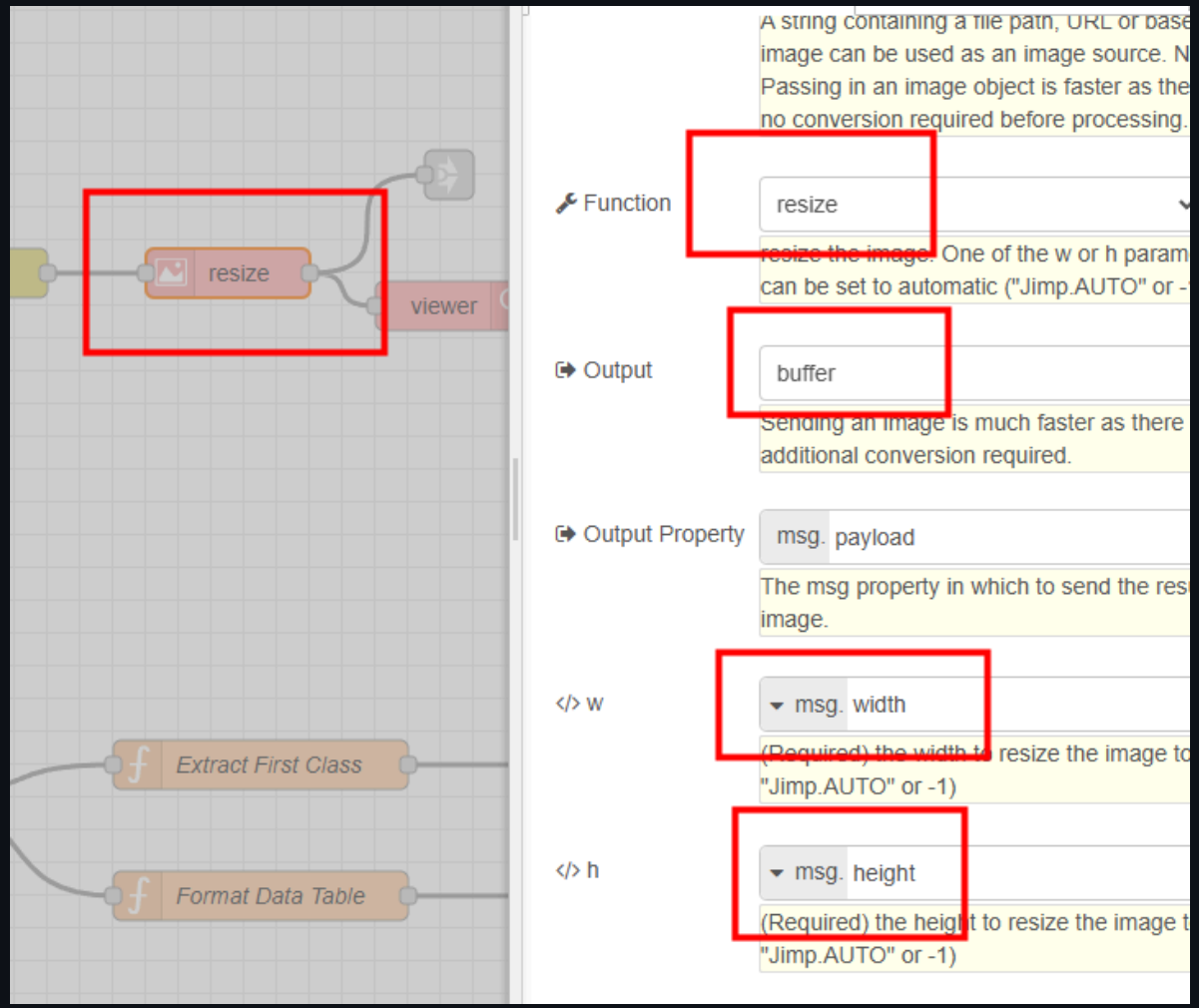
Change output to `buffer`.

Change Node Configuration

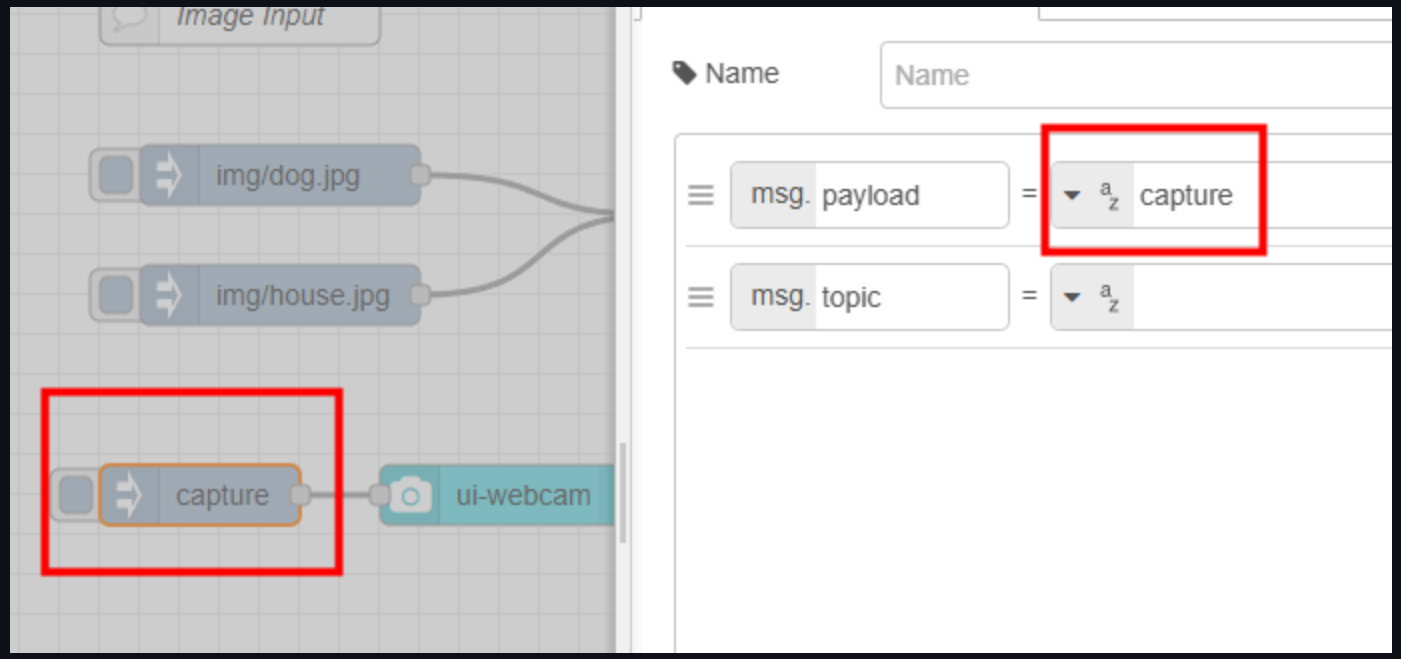


Setting width and height options.

Image Node Configuration (Resize)

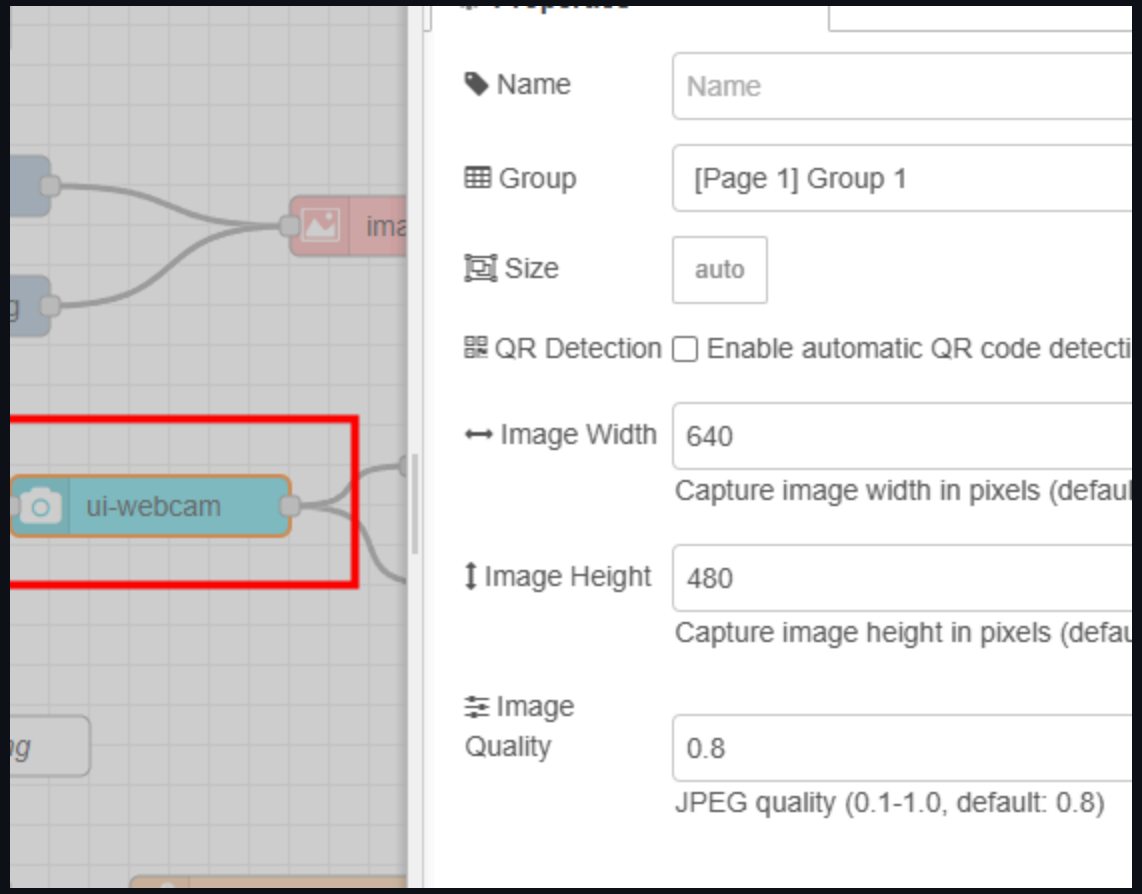


`Inject` Node Configuration (Capture Image)



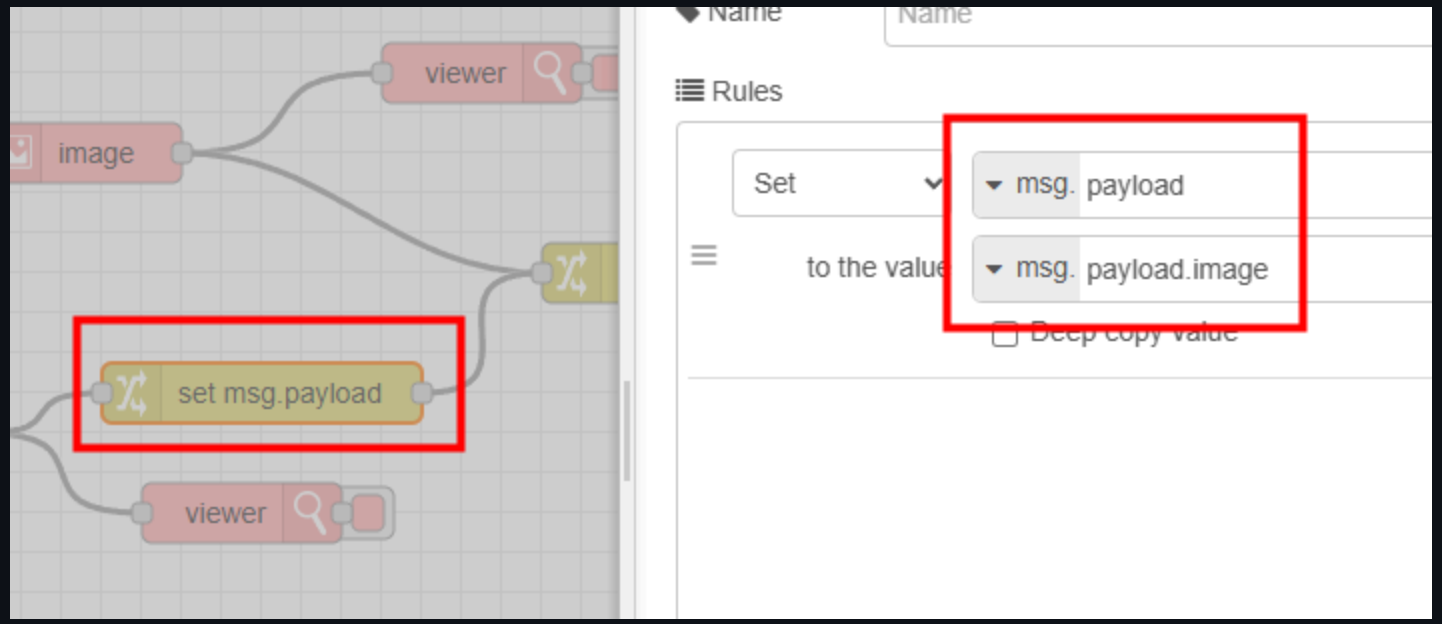
Set payload to `capture` to trigger webcam capture.

`UI-Webcam` Node Configuration



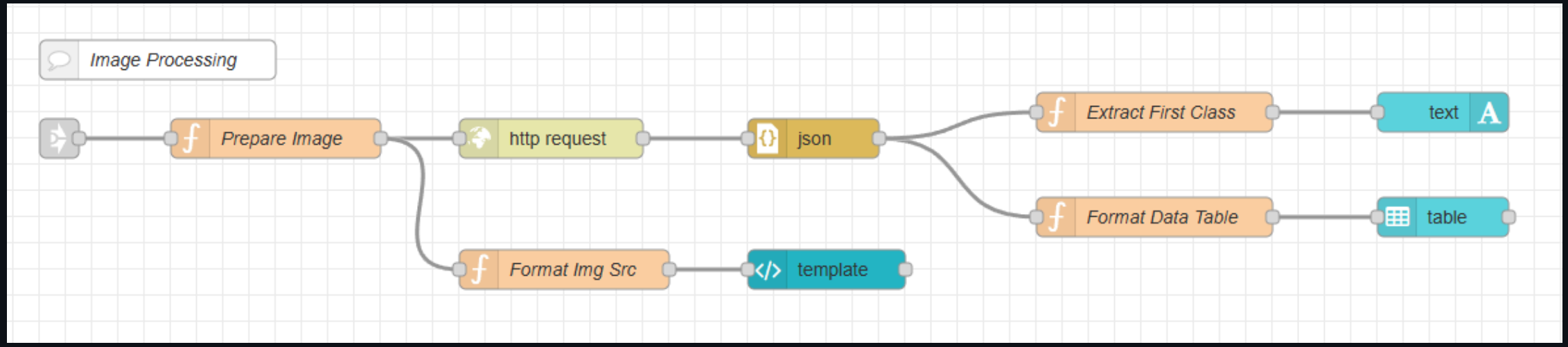
Leave default settings.

`Change` Node Configuration



Extract `image` field from webcam output.

Image-Processing Flow



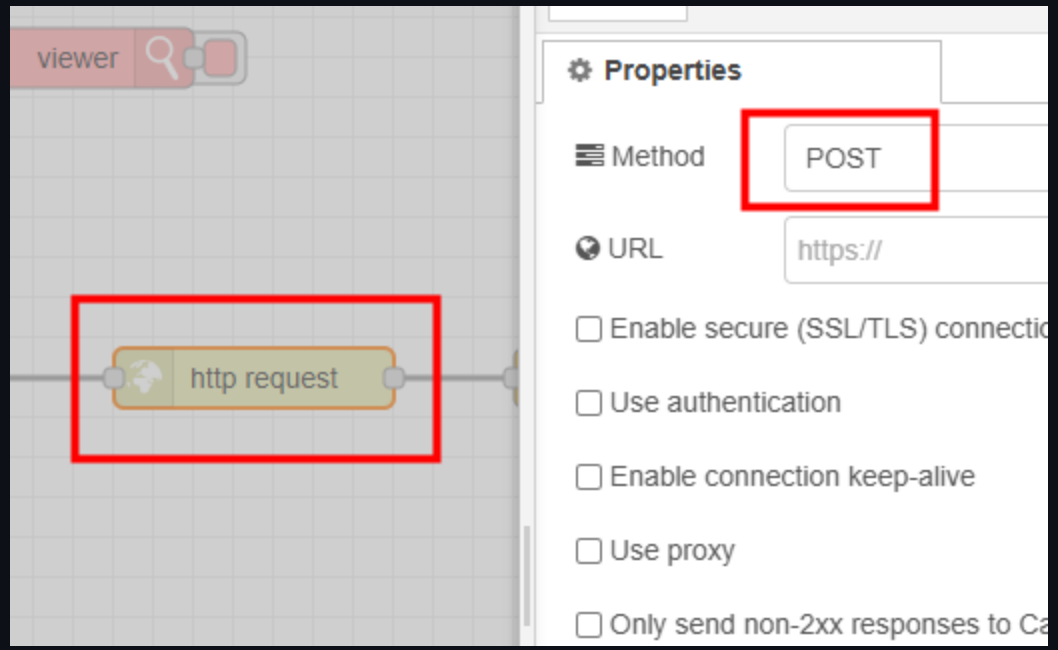
Note

- I attempted to use `complete` node to link payload from `image` node (resize), but it did not get triggered as expected.
  - This might be due to the way `image` node works internally. (Probably not calling `node.done()` function)
- I will use `link` nodes to connect the two flows instead.

`Prepare Image` Function Node

```
msg.payload = { imageEncoded: Buffer.from(msg.payload).toString("base64") };
msg.headers = { "Content-Type": "application/x-www-form-urlencoded" };
msg.url = "http://localhost:3003/upload_base64"; // ML server endpoint
return msg;
```

`HTTP Request` Node Configuration



`Extract First Class` Function Node

```
const predictions = msg.payload?.predictions;
if (Array.isArray(predictions)) {
  msg.payload = predictions[0]?.class ?? "No Class Name";
} else {
  msg.payload = "No Class";
}
return msg;
```

`Format Data Table` Function Node

```
const predictions = msg.payload?.predictions;
if (!Array.isArray(predictions)) {
  msg.payload = [];
  return msg;
}

const dt = new Date();
const datestring = dt.toLocaleDateString();
const timestring = dt.toLocaleTimeString();
msg.payload = predictions.map((d) => ({
  Class: d["class"],
  Score: `${Math.round(d["score"] * 100)}%`,
  Time: `📅${datestring} ⏰${timestring}`,
}));
return msg;
```

`Format Img Src` Function Node

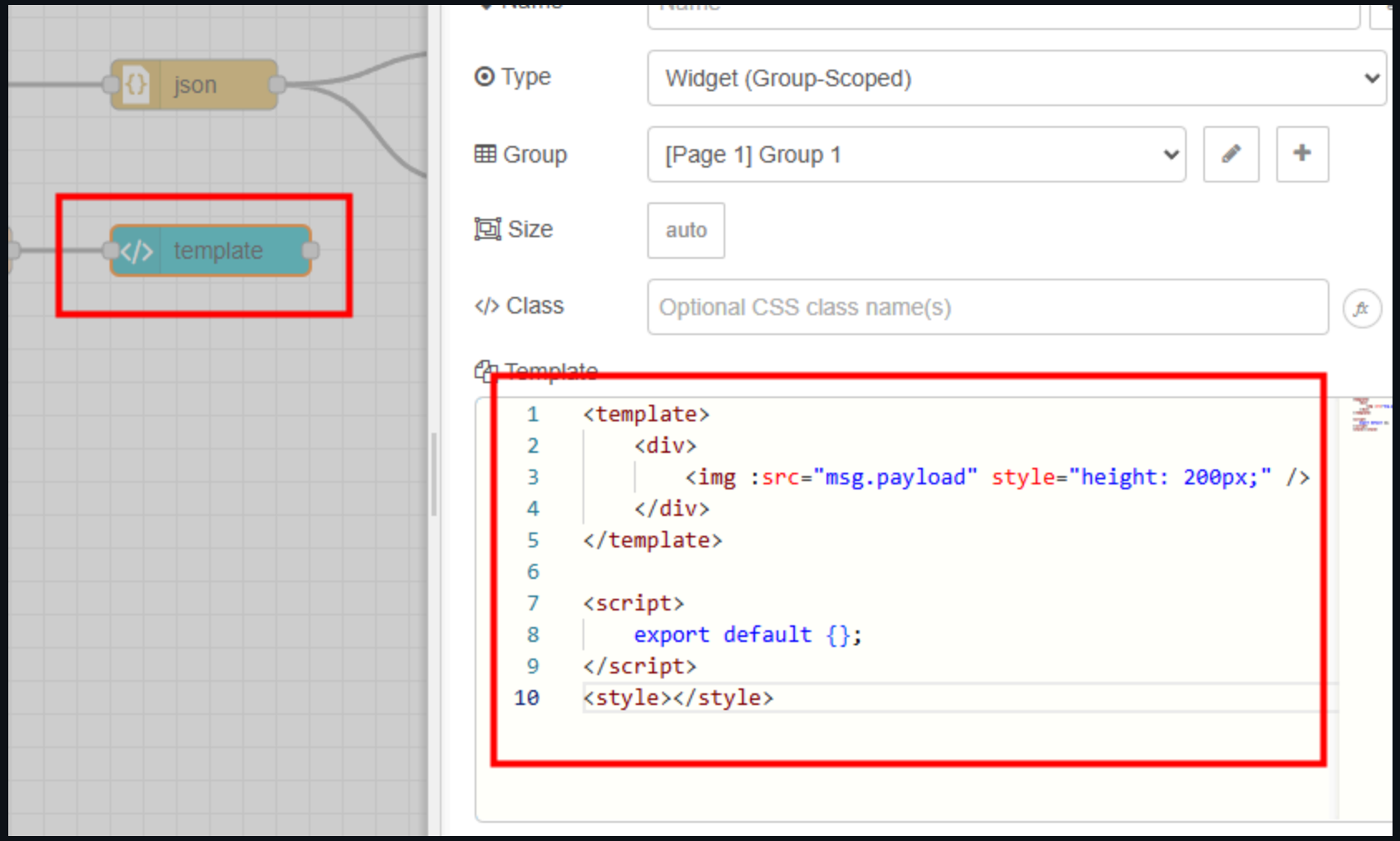
```
msg.payload = "" + msg.payload.imageEncoded;
return msg;
```

`UI-Template` Node Configuration

```
<template>
  <div>
    <img :src="msg.payload" style="height: 200px;" />
  </div>
</template>

<script>
export default {};
</script>
<style></style>
```

Display image from base64 string.


json
template
Type
Widget (Group-Scoped)
Group
[Page 1] Group 1
Size
auto
Class
Optional CSS class name(s)
Template
<template>

</template>
<script>
export default {};
</script>
<style></style>

Train your own model

- Teachable machine
- Create an image classification model.
- Download model and extract to `ml-express/public/model` folder.